

SMART BUILDINGS — MASTER IDEATION DOCUMENT

Dynamic Building Energy Flexibility Engine

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The idea in one sentence

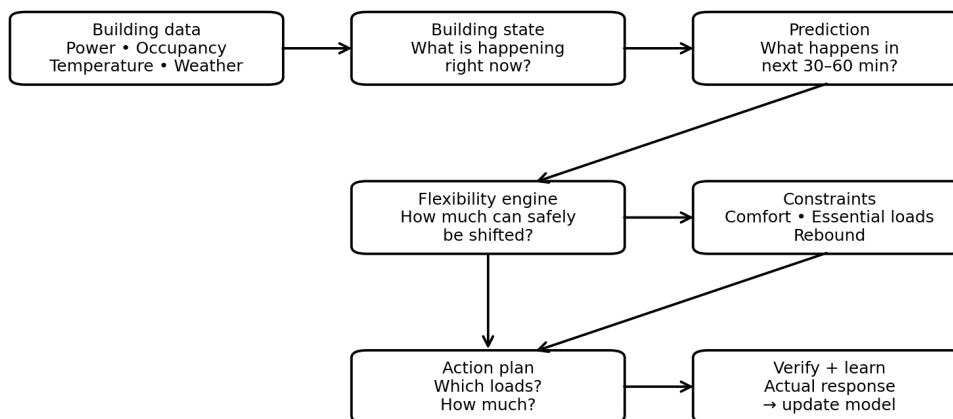
Make a building aware of its own real-time energy flexibility: how much electrical demand can be reduced or shifted, from which loads, for how long, while preserving comfort and accounting for rebound.

1. Executive Summary — Everyone Should Understand This

A building's total electricity load is not equal to its flexible load. At 2 PM a building may consume 500 kW, but only part of that 500 kW may be safely reducible. The flexible amount changes continuously with occupancy, weather, indoor temperature, HVAC state, equipment requirements and what the building has already done.

Our system continuously estimates this changing flexibility. When a grid operator or facility manager asks for a specific reduction, the system tests possible actions and returns a feasible plan rather than applying an arbitrary percentage reduction.

Dynamic Building Energy Flexibility Engine — End-to-End



2. Core Terms — What Each Word Means

Term	Meaning in our project
Current load	Electricity the building is using now.
Flexible load	Demand that can be reduced, delayed or shifted without violating defined constraints.
Flexibility	The amount of flexible demand available at a particular time and for a particular duration.
Comfort constraint	The acceptable indoor operating range used to prevent harmful/undesirable HVAC reduction.
Essential load	Equipment/processes that cannot be interrupted for the scenario.
Rebound	Extra demand after a flexibility event as the building/equipment recovers.
Flexibility envelope	The feasible combinations of reduction amount and duration under current conditions.

3. Exact Problem We Are Solving

Problem statement

Buildings can measure how much electricity they consume, but they often cannot determine how much of that demand is safely shiftable at a given moment. A static reduction rule can cause discomfort, operational disruption or rebound. We need a dynamic way to estimate actionable flexibility under changing building conditions.

- Crowded room → lower HVAC flexibility.
- Empty room → higher HVAC flexibility.
- Critical laboratory equipment → little/no flexibility.
- Weather changes → thermal flexibility changes.
- Occupancy changes → thermal demand changes.
- Previous load reduction → recovery/rebound changes.

4. Proposed Solution — Full System

The Dynamic Building Energy Flexibility Engine is a software layer that sits above building data/control systems. It reads the building state, predicts near-term behaviour, evaluates possible control actions and produces an actionable flexibility estimate.

Stage	System does	Output
1. Sense	Reads power, occupancy, temperature, weather and load state	Current building state
2. Predict	Predicts near-term thermal/load behaviour	Expected demand/state
3. Generate actions	Creates possible load-shifting/reduction actions	Candidate control plans
4. Constrain	Checks comfort, essential loads and operating limits	Feasible plans
5. Evaluate	Calculates energy, peak, duration and rebound	Plan scores
6. Decide	Selects best feasible response	Recommended action
7. Verify	Measures simulated/real response	Actual vs predicted
8. Learn	Updates model/estimate	Improved future flexibility

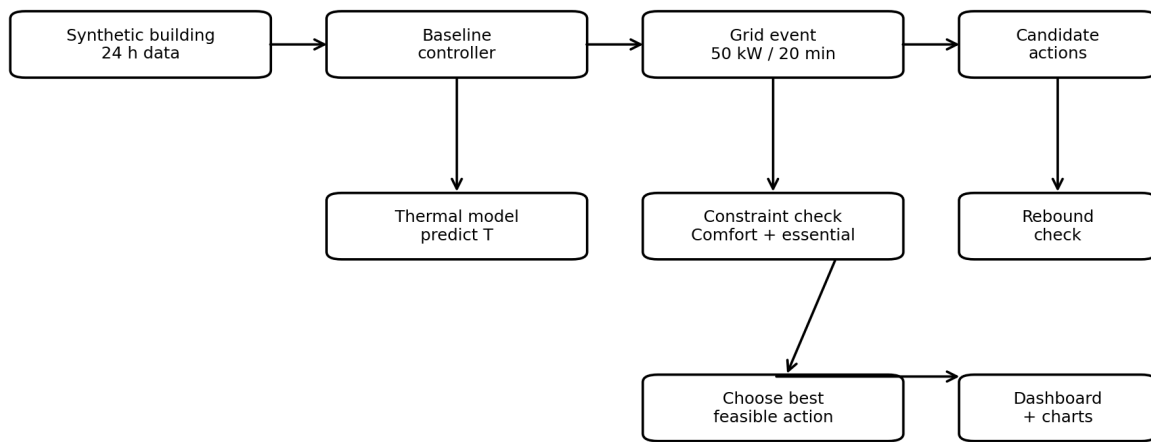
5. The Online Software Simulation

What we will actually build in two days

A Streamlit web app with a simulated building. It will not pretend to be a deployed BMS. It will demonstrate the algorithm and its behaviour under different conditions.

Recommended stack: Python + NumPy + Pandas + Plotly + Streamlit.

Two-Day Software Simulation Flow



Simulation inputs

- 24-hour time series at 5-minute intervals.
- Outdoor temperature/weather profile.
- Occupancy schedule with uncertainty.
- Indoor temperature.
- Base electrical load.
- HVAC power/capacity.
- Lighting load.
- Optional EV charging or battery load if time permits.
- Demand-response request: requested kW reduction + duration.

Simulation outputs

- Current load.
- Available flexibility in kW.
- Safe duration.
- Recommended contributing loads.
- Comfort risk.
- Expected rebound.
- Baseline vs proposed energy.

- Baseline vs proposed peak demand.
- Indoor temperature and comfort chart.
- Confidence/uncertainty indicator.

6. Scenario 1 — College Building

Primary demo scenario. This is easy to explain and gives the evaluator a realistic demand-response story.

Condition	Example
1:30 PM	70% occupied; hot outside; normal classes
2:00 PM	Large lecture starts; occupancy rises to 95%
2:20 PM	Grid/facility request: reduce 50 kW for 20 minutes
System decision	Estimate current flexibility; do not blindly reduce 10%
Possible result	HVAC + lighting + optional EV provide feasible reduction
After 2:40 PM	System checks rebound/recovery and updates estimate

Illustrative result only

Requested: 50 kW / 20 min → estimated available: 56 kW → recommended: HVAC 32 kW + lighting 8 kW + EV 12 kW → comfort risk low → expected rebound 10 kW. Final numbers must come from the simulation.

7. Scenario 2 — Office Building

Second demo scenario. The same building can have very different flexibility at different times, proving that flexibility is dynamic rather than a fixed percentage.

Time	State	What changes
11:00 AM	40% occupancy; mild conditions	Higher potential flexibility
12:00 PM	Meeting rooms fill	HVAC flexibility decreases
1:00 PM	Weather/solar conditions change	Thermal response changes
2:00 PM	Grid request arrives	Engine recalculates from current state
After event	Building recovers	Rebound is measured

8. Technical Approach

Use a simple transparent thermal model rather than a full digital twin. The model only needs to reproduce realistic directional behaviour: higher outdoor heat/occupancy increases cooling demand; lower HVAC output increases indoor temperature; previous pre-cooling or load reduction changes later demand.

$$T_{next} = T + \Delta t / C \times [UA(T_{out} - T) + Q_{internal} + Q_{solar} - Q_{HVAC}]$$

Then use candidate control trajectories over a short horizon. Reject any candidate that violates the comfort or essential-load constraints. Among feasible candidates, choose the one that meets the grid request with the lowest combined energy, rebound and risk.

9. What the Online Demo Screen Should Look Like

UI component	Display
KPI 1	Current Load: 500 kW
KPI 2	Available Flexibility: 52 kW

KPI 3	Safe Duration: 20 min
KPI 4	Comfort Risk: Low
KPI 5	Expected Rebound: 11 kW
Control	Requested reduction slider: 10–80 kW
Control	Event duration: 5–60 min
Control	Occupancy/weather scenario selector
Chart	24-hour baseline vs proposed load
Chart	Indoor temperature vs comfort band
Chart	Flexibility contribution by load
Panel	Recommended action

The “wow” interaction

Move the requested reduction from 30 kW → 50 kW → 70 kW. The system should change the feasible answer and explain when the request becomes infeasible or high-risk.

10. Existing Approaches We Need to Know

Approach	What it generally does	Our distinction
Occupancy-driven HVAC/lighting	Controls loads from occupancy/schedules	We estimate total actionable flexibility for a demand request.
BMS dashboards	Monitor energy/equipment/building state	We convert state into a flexibility decision.
Thermal pre-cooling	Shifts cooling in time	It becomes one flexibility mechanism, not the whole project.
Demand response	Reduces/shifts loads during grid events	We estimate whether the building can reliably provide a requested response.
MPC / predictive control	Optimizes future control actions	We use prediction/optimization as part of a broader flexibility engine.

Do not claim these technologies do not exist. The research task is to identify a practical gap around adaptive, low-sensing, actionable flexibility estimation for existing buildings, and to demonstrate it credibly in simulation.

11. Research Tasks — Exactly What Each Person Must Look Into

Research area	Questions to answer	Owner
Challenge requirements	What exactly must be submitted? What metrics/artefacts are required?	Nandini
Existing Schneider capabilities	What already exists for BMS, HVAC optimization, demand response and analytics?	Nandini
Flexibility research	How do papers define/quantify building flexibility? What methods exist?	Nandini
Indian buildings	What building types, sensing limitations and climate-zone assumptions matter?	Akhilesh
Energy/comfort metrics	What baseline, kWh, kW peak and comfort metrics should we report?	Akhilesh
Retrofit/deployment	Minimum sensors/gateway/control requirements; indicative cost categories; deployment path	Akhilesh
Simulation model	Thermal equations, synthetic data, scenarios and validation checks	You
Optimization logic	Candidate actions, constraints, scoring and rebound calculation	Puneeth

Demo testing	Stress-test scenarios and find failure cases	Puneeth
PPT visuals	Architecture, charts, scenario graphics and final narrative	Akhilesh + Nandini

12. Research Output Format

Every research task should produce a small evidence sheet, not a long report:

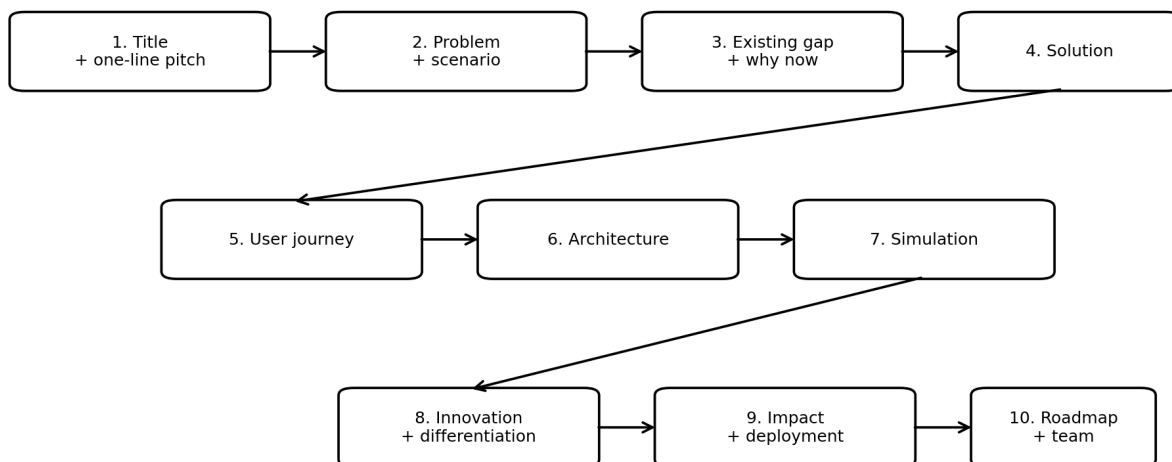
Item	Required output
Finding	1–3 sentence conclusion
Source	Paper/official page/report
Why it matters	How it changes our project
Decision	Keep / change / remove something
Slide use	Which PPT slide it supports

Rule

No one should spend hours collecting random facts. Research only what changes the problem definition, architecture, technical method, differentiation, impact calculation or deployment plan.

13. PPT — Exact 10-Slide Build Plan

Recommended 10-Slide Story Flow



Slide	What to put	Visual
1. Title	Project name + one-line pitch + team	Clean hero architecture/graphic
2. Problem	Dynamic flexibility problem + college scenario	500 kW → unknown flexible portion
3. Gap	Why static rules/BMS/occupancy control are insufficient for a grid request	Comparison diagram
4. Solution	Dynamic flexibility engine + key outputs	End-to-end flowchart
5. User journey	Grid/facility request → assessment → recommendation → action →	5-step journey

	verification	
6. Technical approach	Inputs, thermal model, constraints, optimization, rebound	Architecture + small equation
7. Simulation	Live web demo + baseline/proposed graphs	Screenshots + QR/link if available
8. Innovation	Dynamic, actionable, comfort-aware, rebound-aware, retrofit-oriented	5 innovation blocks
9. Impact/deployment	Energy/peak/comfort metrics, retrofit, climate-zone scalability	Before/after chart + deployment ladder
10. Roadmap + team	48-hour prototype → pilot → multi-load → campus/grid; team roles	Timeline + team cards

14. PPT Rules

- One message per slide.
- Use diagrams and numbers instead of paragraphs.
- Never claim simulated values are field measurements.
- Show at least one live simulation screenshot.
- Show baseline vs proposed results.
- Use the two scenarios to make the idea understandable.
- Keep the detailed equations in speaker notes/appendix if allowed; the main slide should remain readable.
- The first three slides must make the problem obvious before technical details begin.

15. Team Roles — Updated and Balanced

Member	Primary ownership	Concrete deliverables
You	Simulation Lead	Thermal model, synthetic data, flexibility engine, Streamlit app, charts, demo script.
Puneeth	Optimization + Demo Engineering Lead	Candidate control actions, optimization/scoring, rebound logic, testing, second scenario, deployment backup. He should own a substantial technical module, not just assist.
Akhilesh	Impact + PPT Lead	Problem framing, scenarios, quantified impact, retrofit/cost/payback assumptions, PPT storyline and final presentation polish.
Nandini	Research + Architecture Lead	Existing approaches, research gap, Schneider landscape, system architecture, Indian building/climate assumptions, technical references.

Everyone must understand the complete idea. Ownership only determines who is responsible for finishing each part.

16. Two-Day Execution Plan

Time	You	Puneeth	Akhilesh	Nandini
Day 1 AM	Thermal model + data	Optimization formulation	Problem/scenario/impact research	Existing approaches + challenge research
Day 1 PM	Baseline + flexibility engine	Candidate actions + rebound	Retrofit/deployment research	Architecture + research gap
Day 1 Evening	Streamlit UI	Testing + second scenario	Initial PPT skeleton	Sources + technical narrative

Day 2 AM	Polish simulation	Stress-test + edge cases	Impact + PPT charts	Research validation + slide content
Day 2 PM	Demo + screenshots	Demo backup + integration	Final PPT	Final PPT + architecture
Day 2 Evening	Final demo run	Final demo run	Presentation practice	Presentation practice

17. What the Evaluator Should See

- Clear problem: current building load does not reveal safe available flexibility.
- Concrete user: facility manager/grid demand-response event.
- Technical mechanism: state estimation + prediction + constraints + optimization + verification.
- Working software simulation.
- Baseline comparison.
- Quantified simulated impact.
- Comfort preserved in the model.
- Rebound explicitly considered.
- Deployment path to existing buildings.
- Clear extension from one building to campus/portfolio/grid.

18. Risks and How We Avoid Them

Risk	Avoidance
Looks like another HVAC dashboard	Make the core output flexibility + duration + action + rebound.
Numbers look invented	Generate every final number from simulation and label it simulated.
Too broad	Prototype HVAC + one secondary load; architecture remains extensible.
Overcomplicated AI	Use transparent physics + optimization first; ML can be a future enhancement.
No proof	Show baseline vs proposed and at least two scenarios.
Novelty overclaim	Acknowledge existing flexibility/MPC research and state the practical gap.
Demo fails	Keep an offline local Streamlit version and a recorded fallback screen capture.

19. Final Teammate Explanation — Two Scenarios

Scenario A: College asks for 50 kW reduction

At 2:20 PM the building is consuming 500 kW. A grid/facility request asks for 50 kW reduction for 20 minutes. Our system does not simply turn off 10% of the building. It checks occupancy, indoor temperature, HVAC state and other flexible loads. It may determine that 56 kW is feasible, recommend 32 kW HVAC + 8 kW lighting + 12 kW EV, and predict the comfort/rebound consequences. The exact final values come from the simulation.

Scenario B: Same building, different moment

At 11 AM the building is lightly occupied and the weather is milder. The engine might estimate high flexibility. At 2 PM a lecture fills the building and outdoor heat rises; the estimate drops. The same 50 kW request may now be infeasible or may require a different combination of loads. This proves that flexibility is a dynamic property of the building.

20. Final Definition of Done

- A working local/online software simulation.
- Two scenarios.
- Interactive requested-kW control.
- Baseline vs proposed load graph.
- Indoor temperature/comfort graph.
- Flexibility breakdown.
- Rebound calculation.
- Clear architecture diagram.
- 10-slide PPT following the required structure.
- Research evidence for existing approaches and the stated gap.
- Deployment/retrofit plan.
- Team members can each explain the complete system.